

# Samarpartap Passey

Full-Stack Engineer · React · TypeScript · Next.js · Node · AI/LLM Systems

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## PROFESSIONAL SUMMARY

Full-stack engineer who ships products end to end, from React front-ends to Node back-ends and APIs. I founded Sage, a voice AI interview coach, and built it solo from architecture to a live deployed product, owning every layer. I care about execution and shipping working software, with a strong AI/LLM edge from building agentic systems and data pipelines. Comfortable owning features and driving them from idea to production.

## TECHNICAL SKILLS

### Programming Languages

TypeScript, JavaScript, Python, C#

### Frontend

React, Redux, Next.js, TailwindCSS, Responsive Design

### Backend

Node.js, Express.js, RESTful APIs, JWT Authentication, ASP.NET Core MVC

### Databases

PostgreSQL, MongoDB, SQL Server, MySQL, Supabase

### Cloud & DevOps

Git, GitHub Actions, CI/CD, Azure, Vercel, Agile/Scrum

### AI & LLMs

GPT-4.1, Claude, RAG, Agentic Systems, LangChain, LangSmith, Vapi, FAISS (vector search), LLM-as-judge evaluation, prompt engineering, OpenAI API

## PROJECTS

### Sage - Agentic AI Interview Coach

Tech Stack: Next.js · TypeScript · Vapi · GPT-4.1 · Claude (Haiku + Sonnet) · Supabase · Clerk · Vercel

- Founded and built Sage solo, owning every layer from the architecture down to the React interface, and took it from nothing to a live, deployed product heading into beta.
- Designed a two-model interview flow where GPT-4.1 runs the live voice conversation through Vapi while Claude generates the structured answer evaluation and final assessment, choosing each model for what it does best.
- Built the interview to generate a tailored plan from the candidate's resume and the specific job description, so the questions match the actual role instead of being generic.
- Wired up input validation, token-limit handling, and fallback logic so the system fails cleanly instead of looping when a model response comes back outside the expected range.
- Cut voice response latency to under 500ms and tracked down cross-browser speech recognition failures on Safari and mobile that only showed up in production.
- Shipped on Vercel with a CI/CD pipeline, Clerk authentication, and session data persisted to Supabase.

[Github](#)

[Live Demo](#)

### Finflow - AI-Powered Financial Analytics Platform

Tech Stack: React · Next.js · TypeScript · Node.js · Express.js · MongoDB · Vercel

- Built a full-stack fintech platform end to end, from the React front-end through Node and Express APIs to a MongoDB backend, handling bank data ingestion and anomaly detection across multiple statement formats.
- Designed RESTful APIs for parsing, categorizing transactions, and flagging anomalies, and wrote Jest unit and integration tests covering the API routes and data validation.
- Built JWT authentication with role-based access control to protect multi-user financial data.
- Deployed on Vercel with a CI/CD pipeline, owning the work from planning through production.

[Github](#)

[LIVE DEMO](#)

### Hybrid RAG Pipeline with LangSmith Observability

Tech Stack: Python · LangChain · FAISS · BM25 · CrossEncoder · LangSmith

- Built a hybrid RAG pipeline in Python with FAISS vector search and BM25 retrieval, plus CrossEncoder reranking, and wrote a custom LLM-as-judge evaluator to benchmark answer correctness and catch regressions across prompt and retrieval changes.
- Integrated LangSmith tracing and offline evaluation across every step (retrieval, prompt assembly, token usage, latency) for full model performance monitoring and visibility into system behavior at inference time.
- Built validation and guardrails to catch retrieval failures, malformed model output, and cost overruns before results reached the user.

[Github](#)

## WORK EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                     |                      |
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| <b>Walmart</b>                                                                                                                                                                                                                                                                                                                      | St. Catharines, ON   |
| General Merchandise Associate                                                                                                                                                                                                                                                                                                       | July 2025 to Present |
| <ul style="list-style-type: none"><li>• Balanced part-time shifts while independently designing and shipping production software outside work hours, maintaining consistent delivery on both.</li><li>• Worked in a fast-paced team environment requiring clear communication and reliable execution under time pressure.</li></ul> |                      |

## EDUCATION

|                                 |                                                 |
|---------------------------------|-------------------------------------------------|
| <b>Niagara College</b>          | Advanced Diploma                                |
| Computer Programming & Analysis | Welland, ON • Sep 2023 - Graduated January 2026 |